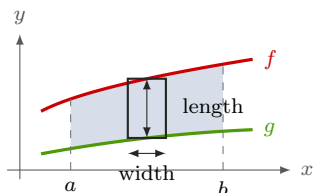


L4.1a Areas Between Curves

Crossing curves — what you need to know

AREA BETWEEN TWO CURVES



$$A = \int_a^b f(x) - g(x) dx$$

f and g continuous on $[a, b]$, with f on top

$$\text{length} = f(x_i^*) - g(x_i^*), \quad \text{width} = \Delta x$$

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n (f(x_i^*) - g(x_i^*)) \Delta x$$

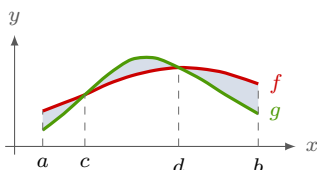
“each rectangle is as tall as the gap between the curves and as wide as Δx ”

Sometimes written $A = \int_a^b y_T - y_B dx$ — the same integrand, named off the picture.

The four steps

- ① **Sketch both curves and shade the region.**
- ② **Get the limits.** Given, or solve $f(x) = g(x)$ — they are the x -coordinates.
- ③ **Test a sample point** to decide which curve is on top. Evaluate both there and compare.
- ④ **Integrate top minus bottom.** If the curves trade places, one integral per region.

CROSSING CURVES



$$A = \int_a^c f - g dx + \int_c^d g - f dx + \int_d^b f - g dx$$

The integrand is always $|f - g|$ — splitting at the crossings is how you compute it without writing the bars.

WATCH OUT

- ✗ ~~f is on top because it looks bigger~~ → settled by testing a point in the interval
- ✗ ~~one integral straight across a crossing~~ → one integral per region — they traded places
- ✗ ~~the limits are the y -coordinates~~ → the x -coordinates — you are integrating dx
- ✗ ~~the area came out negative~~ → you subtracted in the wrong order
- ✗ ~~under the axis, so add that piece back~~ → top minus bottom already measures the gap